Jawad Adit Chowdhury

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Professional Summary

Computer Science and Engineering graduate from BRAC University building full-stack web apps (React, Node.js, MongoDB) and applying deep learning to medical imaging (PyTorch, nnU-Net). Delivered a team-designed nnU-Net v2 output head for pediatric brain tumor segmentation (five-fold cross-validation, 257 MRI cases); a deployed AI subscription and finance platform built with a four-person team; and a sustainability tracker with personalised Gemini API advice. Experienced in REST APIs with authentication and ownership checks, interactive React dashboards, and controlled ML experiments with statistical testing.

Education

- **BRAC University**
  Bachelor of Science in Computer Science and Engineering; CGPA: [3.18 / 4.00] *2022 – 2026*

Technical Skills

- **Programming Languages**: JavaScript, TypeScript, Python
- **Frontend**: React, Tailwind CSS, Bootstrap, Chart.js, Recharts
- **Backend & Data**: Node.js, Express.js, MongoDB, Mongoose, REST APIs
- **Machine Learning**: PyTorch, nnU-Net, medical image segmentation, statistical testing
- **Tools**: Git/GitHub, Postman, Figma, Vercel, pytest, draw.io

Projects

- **Automated 3D Segmentation of Pediatric Brain Tumors**
  PyTorch, nnU-Net v2, Python, BraTS-PEDs 2025
  - Worked with team members to design HSRD, a hybrid sigmoid-softmax output head for nnU-Net v2 that makes tumor sub-regions mutually exclusive via a shared softmax, ground-truth-masked loss, and an inference gate.
  - Evaluated against 3 baselines with five-fold cross-validation on 257 MRI cases using Dice, HD95, lesion-wise metrics, and paired Wilcoxon tests with Holm correction.
  - Removed sibling-label conflicts that appeared in 48 of 52 baseline validation cases and raised correctly empty ET-free predictions from 27 to 53 of 86 cases.
- **Stackwise: AI Subscription Optimizer & Personal Finance Tracker**
  React, Node.js, Express, MongoDB, Tailwind CSS, Gemini API
  - Built the subscription tracking module: a secured Express/MongoDB REST API (CRUD with per-user ownership checks) and a React drawer form with capability tags, trial alerts, and optional checkout.
  - Built a client/project profitability dashboard comparing revenue, costs, and margins per client, with Recharts category breakdowns.
  - Built a tax report generator with downloadable PDF export and a Gemini-powered scan that flags missed possible tax deductions.
  - Designed the subscription tracker screens in Figma. Deployed on Vercel (Lighthouse: 85 performance, 95 accessibility).
- **EcoTrack: Personal Sustainability Tracker**
  React, Node.js, Express, MongoDB, Chart.js, Gemini API
  - Built a MERN app (MVC) where users log eco-friendly and harmful activities to earn an Eco Score and a per-category carbon footprint estimate.
  - Implemented goals with progress tracking, badges and streaks, Chart.js trend and category charts, and daily reminders at a configurable time.
  - Integrated the Gemini API to generate personalised advice and suggested activities from each user's logged history.

Training & Extracurriculars

- **Certifications**: SBCRD – Stata (2022), Sarbojonkotha - Research Topic and Methodology (2024)
- **Extracurriculars**: BRACU Multicultural Club - General Member(2023-2025), DUNMUN - Delegate(2022), JUMUN - Delegate(2024), Scout - Patrol Leader

Personal Details

- **Spoken Languages**: English (*Fluent*), Bengali (*Fluent*), French (*Basic*)
- **Date of Birth**: 05th November 2002